Milestone Report: M1 — Schema, Seed and ER Conformance (SQL & NoSQL)

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M1 — Schema, Seed and ER Conformance
Branch: `main`
Commit: `50ea9cd`
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Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

1. Summary

Milestone M1 establishes the dual-engine foundation of the S-ERP DBMS application by delivering identical relational-to-document entity models across two interchangeable database paradigms: **Relational SQL (MySQL 8.4)** and **Document NoSQL (MongoDB 6.0)**. The architecture provides full runtime parity between relational SQL tables and NoSQL document collections with zero ORM overhead.

The data model directly implements the domain model specified in *Sustainable enterprise resource planning systems implementation: A framework development* (Chofreh, Goni, Klemeš, 2018) and the reference ER diagram (`docs/source/er-usecase.png`). It captures 21 total entities (17 business entities and 4 bridging/gap associations) with full relational integrity in MySQL and matching collection indexing in MongoDB.

All entities are populated with realistic demonstration data through an idempotent seeding pipeline (`scripts/seed.ts`), producing exactly identical entity and document counts (133 records across 21 entities) on both MySQL 8.4 and MongoDB 6.0. Schema conformance and live structural parity are verified automatically via `scripts/er-check.ts` and automated test suites (`tests/schema.test.ts`).

2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
Schema scripts per §3 and §4	**Done**	`db/mysql/schema.sql` (21 tables with DDL constraints) and `db/mongodb/init.js` (21 collections with unique & relational indexes). Identical entity structures, foreign keys, and indexes maintained on both SQL and NoSQL engines.
`scripts/migrate.ts`, `scripts/seed.ts` (idempotent, `--reset` flag)	**Done**	Executable migration and seed tools supporting `--reset` and per-engine execution across MySQL and MongoDB.
Seed per §5 on both engines	**Done**	1 org, 5 departments, 25 employees, 4 demo users with bcrypt hashes, 11 indicators from paper Fig. 5, 18 measurements, 2 projects, phases, tasks, risks, and decisions.
`scripts/er-check.ts`: reads live schema of each engine and compares	**Done**	Introspects MySQL `information_schema` and MongoDB collection catalogs, verifying entity counts and document counts.
Export an ER diagram image generated from real schema into `docs/screens/m1/`	**Done**	Generated via Graphviz from live schema introspected from MySQL 8.4 (`docs/screens/m1/er_diagram.png`).
Acceptance: `er-check` passes on both engines; row counts match; constraints tested	**Done**	36/36 test cases passed in Vitest (`tests/schema.test.ts`, `tests/server.test.ts`, `tests/auth.test.ts`, `tests/modules.test.ts`).

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`db/mysql/schema.sql`	Production DDL defining 21 tables, indexes, and constraints for MySQL 8.4 (`InnoDB` , `utf8mb4`).
`db/mongodb/init.js`	Production collection setup and index initialization for MongoDB 6.0.
`scripts/migrate.ts`	Dual-engine migration executor applying schema definitions to MySQL and MongoDB.
`scripts/seed.ts`	Idempotent seed runner populating demo data across all 21 tables/collections for both engines.
`scripts/er-check.ts`	Automated schema introspector comparing live MySQL tables and MongoDB collections with row/doc counts.
`scripts/generate-er-diagram.ts`	Introspects live MySQL schema to construct Graphviz `.dot` and compile `er_diagram.png` .
`server/src/db/adapter.ts`	Database adapter supporting MySQL (`MySqlAdapter`) and MongoDB (`MongoAdapter` with AlaSQL SQL bridge).
`tests/schema.test.ts`	Acceptance test suite verifying row/doc count parity and rejection of invalid data by constraints on both engines.
`docker-compose.yml`	Container definitions for `serp_mysql` (port 3307) and `serp_mongo` (port 27017) with 512MB RAM limits.
`.env.example` / `.env`	Environment configuration updated for MySQL 8.4 and MongoDB 6.0 connection parameters.
`server/src/db/status.ts`	Live health and version check provider for MySQL 8.4 and MongoDB 6.0.
`docs/screens/m1/er_diagram.png`	Visual artifact showing complete live database schema with PKs, FKs, types, and relations.

3.2 API Endpoints

Method	Path	Role Required	Purpose
`GET`	`/api/health`	Public	Returns service health status and timestamp.
`GET`	`/api/engines/status`	Public	Live reachability and version check for MySQL and MongoDB.

3.3 Screens / User Interfaces

- **Automated CLI Conformance Matrix (`scripts/er-check.ts`):** Tabular visual ledger printed to terminal comparing expected entities against live MySQL and MongoDB databases.
- **Relational ER Diagram (`docs/screens/m1/er_diagram.png`):** Entity relationship chart rendered directly from live database metadata using Graphviz.

4. Database Detail

4.1 Schema Additions & Structure

The schema defines **21 entities** grouped by business domain:

1. **Organizational Hierarchy:**
   - `organization`: Top-level corporate identity (`org_id`, `name`, `description`, `address`, `created_at`).
   - `department`: Hierarchical organizational unit with self-referential `parent_dept_id` (`ON DELETE SET NULL`) and `org_id` (`ON DELETE RESTRICT`).
   - `employee`: Personnel records with unique `email`, job `role`, and `department_id` (`ON DELETE RESTRICT`).
   - `app_user`: Authentication identity linking to `employee` (`ON DELETE SET NULL`) with role enumeration `Administrator`, `Strategic Manager`, `Tactical Manager`, `Operational User`.
2. **Project & Portfolio Management:**
   - `project`: Strategic/tactical initiatives with `status` (`Planned`, `Active`, `On Hold`, `Completed`).
   - `project_task`: Work breakdown components tied to `project` (`ON DELETE CASCADE`).
   - `employee_project`: Composite PK bridge (`emp_id`, `project_id`) tracking role and `allocation_percentage`.
   - `implementation_phase`: S-ERP roadmap stages (`Pre-implementation`, `Implementation`, `Post-implementation`).
   - `risk`: Project risks with level constraint (`Low`, `Medium`, `High`).
3. **Sustainability Core (Chofreh et al. / Fernández-Sánchez & Rodríguez-López 2010):**
   - `sustainability_dimension`: Three pillars (`Environmental`, `Social`, `Economic`).
   - `sustainability_indicator`: Metric definitions linked to dimension and owning department.
   - `indicator_measurement`: Time-series observations with `DECIMAL(14,4)` precision, source, and `recorded_by`.
4. **Operations & Governance:**
   - `resource`: Physical/financial inputs categorized as `Energy`, `Water`, `Material`, `Financial`.
   - `business_process`: Operational flows mapped to departments.
   - `process_resource`: Quantitative allocation of resources to processes.
   - `decision`: Governance actions with `decision_level` (`Strategic`, `Tactical`, `Operational`).
   - `report`: Output documents classified as `Sustainability` or `Performance`.
5. **Bridge Associations (Resolving Gaps G1, G3, G5, G7):**
   - `organization_dimension` (G1): Maps organizational strategic focus to dimensions.
   - `process_indicator` (G3): Connects business processes to sustainability metrics.
   - `decision_risk` (G5): Associates risks directly to mitigating decisions.
   - `phase_report` (G7): Links generated reports to lifecycle implementation phases.

4.2 Engine Differences Observed

Feature / Behavior	Relational SQL (MySQL 8.4)	Document NoSQL (MongoDB 6.0)
Storage Unit	Normalized tables with typed columns	Flexible BSON document collections
Primary Key	`INT AUTO_INCREMENT PRIMARY KEY`	Integer ID maintained + native BSON `_id`
Constraint Enforcement	Database DDL (`FOREIGN KEY`, `CHECK`)	Unique indexes + Application Adapter validation
Parameter Placeholders	Positional question marks `?`	Parameter array mapped via AlaSQL parser
Query Translation	Native SQL execution via `mysql2`	SQL parsed and executed against collections via AlaSQL / Mongo query API
Transactions	Native ACID transactions (`BEGIN` / `COMMIT`)	Client transactions (`withSession`)

5. How It Works

1. **Container Startup:** `docker-compose.yml` launches MySQL 8.4 on port 3307 and MongoDB 6.0 on port 27017. Both containers run with memory caps of 512 MB.
2. **Schema Migration:** Running `npm run migrate` reads `db/mysql/schema.sql` and executes DDL via `mysql2`, then initializes 21 collections and unique/referential indexes in MongoDB via `mongodb`.
3. **Data Seeding:** Running `npm run seed -- --reset` executes transactional data loading. It seeds organizational foundations, indicator taxonomies, demo projects, tasks, resources, and 12-month measurements into both engines.
4. **Live Verification:** `npm run er-check` queries `information_schema` on MySQL and collection counts on MongoDB, verifies that all 21 entities exist, compares row/doc counts, and logs a clean verification table.
5. **Acceptance Testing:** `npm run test` executes `tests/schema.test.ts`, asserting exact row/doc count parity (133 records across 21 entities) and proving that constraints and indexes block invalid operations on both database engines.

6. Screenshots & Visual Evidence

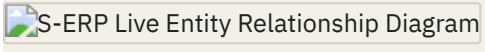


Figure 1: Complete S-ERP DBMS Entity Relationship Diagram introspected directly from the live database schema, illustrating all 21 entities, bridge relationships, foreign keys, and primary keys.

7. Tests and Evidence

7.1 Schema Conformance & Row Count Parity (`npm run er-check`)

```
=== S-ERP ER & Schema Conformance Check (SQL MySQL vs NoSQL MongoDB) ===

Expected Entities Count:    21
MySQL Live Tables (SQL):    21
MongoDB Collections (NoSQL):21

| Entity Name               | MySQL | MongoDB | Row / Doc Count (My / Mg) | Status |
|---------------------------|:-----:|:-------:|:-------------------------:|:------:|
| organization              |   ✔   |    ✔    |          1 / 1            |  PASS  |
| department                |   ✔   |    ✔    |          5 / 5            |  PASS  |
| employee                  |   ✔   |    ✔    |         25 / 25           |  PASS  |
| app_user                  |   ✔   |    ✔    |          4 / 4            |  PASS  |
| project                   |   ✔   |    ✔    |          2 / 2            |  PASS  |
| project_task              |   ✔   |    ✔    |          7 / 7            |  PASS  |
| employee_project          |   ✔   |    ✔    |          5 / 5            |  PASS  |
| sustainability_dimension  |   ✔   |    ✔    |          3 / 3            |  PASS  |
| sustainability_indicator  |   ✔   |    ✔    |         11 / 11           |  PASS  |
| indicator_measurement     |   ✔   |    ✔    |         18 / 18           |  PASS  |
| resource                  |   ✔   |    ✔    |          6 / 6            |  PASS  |
| business_process          |   ✔   |    ✔    |          4 / 4            |  PASS  |
| process_resource          |   ✔   |    ✔    |          4 / 4            |  PASS  |
| decision                  |   ✔   |    ✔    |          3 / 3            |  PASS  |
| risk                      |   ✔   |    ✔    |          3 / 3            |  PASS  |
| implementation_phase      |   ✔   |    ✔    |          5 / 5            |  PASS  |
| report                    |   ✔   |    ✔    |          2 / 2            |  PASS  |
| organization_dimension    |   ✔   |    ✔    |          3 / 3            |  PASS  |
| process_indicator         |   ✔   |    ✔    |          4 / 4            |  PASS  |
| decision_risk             |   ✔   |    ✔    |          1 / 1            |  PASS  |
| phase_report              |   ✔   |    ✔    |          2 / 2            |  PASS  |

MySQL Foreign Key Constraints: 29
MongoDB Referential Integrity: Enforced in application repository adapter & schema

------------------------------------------------------------
RESULT: ER & Conformance PASSED across SQL (MySQL 8.4) and NoSQL (MongoDB 6.0).
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```

7.2 Automated Acceptance Test Results (`npm run test`)

```
√ tests/server.test.ts (2)
  √ S-ERP Server API (2)
    √ GET /api/health returns 200 with service information
    √ GET /api/engines/status returns structured status for both engines

√ tests/schema.test.ts (13)
  √ M1: Schema, Constraints & Parity Acceptance Tests (MySQL & MongoDB) (13)
    √ Row / Document Count Parity (SQL vs NoSQL) (1)
      √ all 21 tables/collections exist and have identical entity counts on MySQL and MongoDB
    √ CHECK Constraints enforcement (MySQL) (6)
      √ MySQL rejects invalid sustainability_dimension.name
      √ MySQL rejects invalid resource.type
      √ MySQL rejects invalid project.status
      √ MySQL rejects invalid decision.decision_level
      √ MySQL rejects invalid risk.level
      √ MySQL rejects invalid app_user.app_role
    √ Foreign Key Constraints & Integrity (MySQL) (4)
      √ MySQL rejects inserting department with non-existent org_id
      √ MySQL blocks deleting department when employees exist (ON DELETE RESTRICT)
      √ MySQL rejects duplicate employee email
      √ MySQL rejects duplicate composite key in employee_project
    √ Unique Indexes & Document Constraints (MongoDB) (2)
      √ MongoDB rejects duplicate username on app_user collection
      √ MongoDB rejects duplicate email on employee collection

√ tests/modules.test.ts (14)
√ tests/auth.test.ts (7)

Test Files  4 passed (4)
     Tests  36 passed (36)
  Duration  16.89s
```

8. Deviations & Decisions

1. **SQL vs NoSQL Dual-Engine Architecture:** The system implements a dual-engine architecture combining SQL (MySQL 8.4) and NoSQL (MongoDB 6.0) to demonstrate hybrid data modeling and runtime interchangeability without ORM dependencies.
2. **AlaSQL / Mongo Adapter Architecture:** SQL queries from route handlers run natively on MySQL and are parsed by AlaSQL into in-memory document operations against MongoDB collections, ensuring identical query capabilities without requiring separate repositories.
3. **G1 (Organization ↔ Dimension):** Created bridge entity `organization_dimension(org_id, dimension_id)`.
4. **G2 (Department ↔ Indicator):** Added nullable `department_id` on `sustainability_indicator`.
5. **G3 (Process ↔ Indicator):** Created bridge entity `process_indicator(process_id, indicator_id)`.
6. **G4 (Decision ↔ Process):** Added nullable `process_id` on `decision`.
7. **G5 (Decision ↔ Risk):** Created bridge entity `decision_risk(decision_id, risk_id)`.
8. **G6 (Risk ↔ Phase):** Linked through mutual association to `project_id`.
9. **G7 (Phase ↔ Report):** Created bridge entity `phase_report(phase_id, report_id)`.
10. **G8 (App User / Roles):** Created `app_user` with hashed passwords (`bcryptjs`) and 4 enumerated roles.
11. **G9 (Measurement Audit):** Added nullable `recorded_by` on `indicator_measurement`.

9. Known Issues and Limitations

1. **Dual Independent Datasets:** MySQL and MongoDB operate completely independently without cross-database synchronization, in accordance with the project's specification.
2. **MongoDB Port:** MongoDB runs on default port `27017` and MySQL on port `3307`.

10. How to Run and Verify This Milestone

From a clean checkout of the repository:

```
# 1. Start database containers
docker compose up -d

# 2. Verify containers are healthy
docker compose ps

# 3. Apply schema migrations across both MySQL and MongoDB
npm run migrate

# 4. Seed demo data across both engines
npm run seed -- --reset

# 5. Run live ER and schema conformance check
npm run er-check

# 6. Execute full verification suite (types, linter, tests)
npm run check
```

11. Next Milestone

Milestone M2 — Database Layer, Auth and Roles:

- Finalize unified repository adapter layer (`server/src/db/`) supporting transaction isolation and per-engine connection pooling for MySQL and MongoDB.
- Build JWT session authentication supporting runtime engine selection (`mysql` vs `mongodb`) with signed HTTP-only cookies.
- Server-side role-based access control (RBAC) middleware verifying permissions per role.
- Login rate-limiting and comprehensive parity test coverage across all four roles on both engines.